

Introduction and Background of the Project

Image classification is a fundamental challenge in computer vision, with applications ranging from autonomous navigation to environmental monitoring. With the rise of deep learning, convolutional neural networks (CNNs) have become the standard for such tasks, continually evolving in complexity and effectiveness. This project explores the performance of several CNN-based models and modern architectural enhancements using the Intel Image Classification Dataset, a well-labeled dataset containing 25,000 images across six distinct scene categories: buildings, forest, glacier, mountain, sea, and street. The dataset consists of real-world images collected from various natural and urban scenes, with images in RGB format uniformly sized to 150×150 pixels. The study evaluates models both trained from scratch and adapted from state-of-the-art vision backbones. The goal is to systematically compare traditional CNN architectures with more advanced alternatives, such as attention mechanisms, residual learning, and feature pyramids, as well as powerful pre-trained models adapted through transfer learning. By analyzing the strengths and weaknesses of each approach, the project aims to uncover the most effective strategies for real-world scene classification tasks.

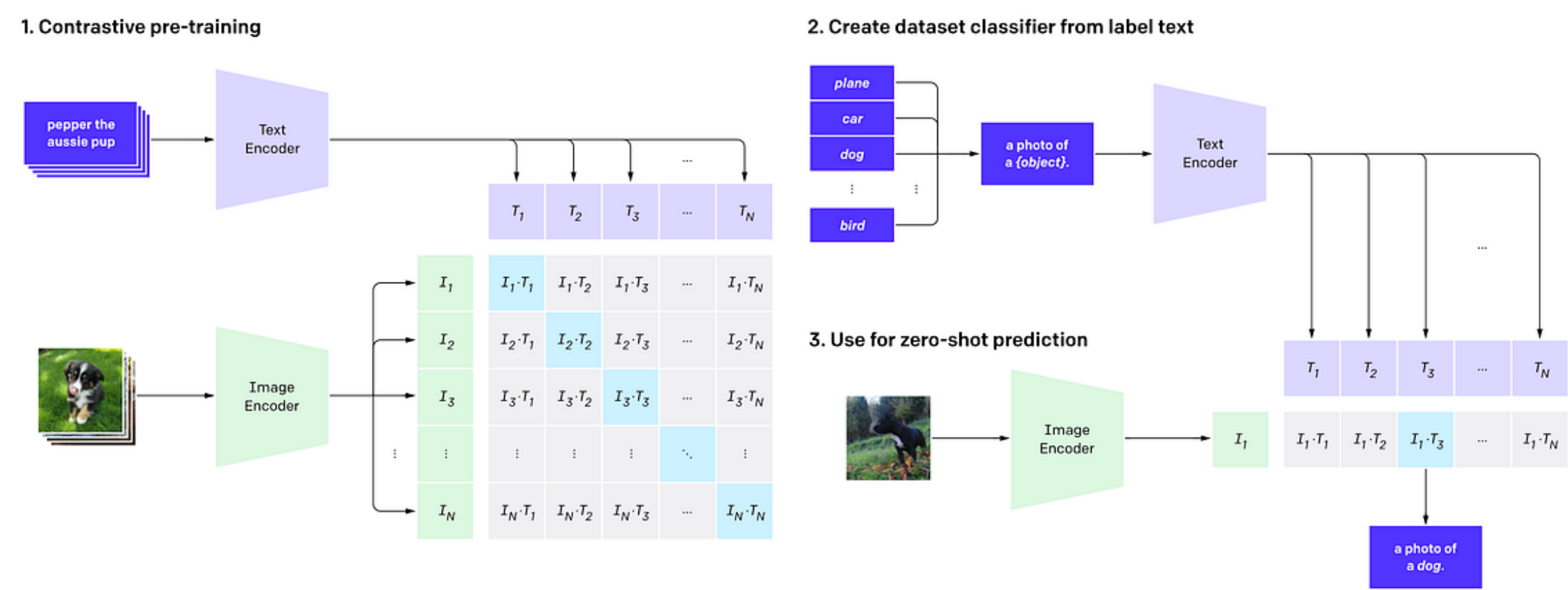


Figure 1. Visualization of images projected into a shared semantic space using OpenAI's CLIP, illustrating the model's ability to cluster semantically similar images together.

Objectives and Research Questions

Objectives: The primary objective is to identify the most effective model for scene recognition by comparing six different deep learning approaches and evaluating their performance using comprehensive metrics.

Research Questions::

- How do different hyperparameter configurations (such as dropout, optimizer, batch normalization) affect the generalization ability of models?
- How do advanced architectural components such as attention blocks, residual blocks, and multi-scale features (FPN) impact the model's performance, compared to models that are stacked up with only convolution and linear layers?
- How much performance advantage can pretrained models achieve compared to the models trained from scratch?
- Can attention-based models (pretrained or from-scratch) improve classification accuracy by focusing only on the most relevant features, despite the presence of inter-class similarity and multi-class visual elements within a single image?

Methodology

Dataset: Intel Image Classification (Six Classes: Buildings, Forest, Glacier, Mountain, Sea, Street).

Classical Architectures: Standard CNNs, Autoencoders, Attention-CNNs, ResNet variants, and Feature Pyramid Networks. Each category includes experiments across depth (3, 5, 7 layers), batch normalization, pooling strategies(Max, Average, None), dropout levels(0.0, 0.1, 0.2, 0.3, 0.5), and optimizers(SGD, Adam, AdamW).

Pretrained Models: CNN-based: EfficientNet-B7, DenseNet-201, ResNet-18/152, ConvNeXt-Large. Transformer-based: Swin-V2-B, CLIP-B32, CLIP-L14. Each was evaluated in frozen and end-to-end finetuned modes.

Multimodal Model: Meta-LLaMa-3.2-11B-Vision-Instruct, zero-shot vs instruction-finetuned on the dataset.

Total Experiments: 270+ controlled runs.

Evaluation Metrics: Validation accuracy, test accuracy, precision, recall, and F1-score. Learning curves for selected models. Boxplot summaries for classical architecture families.

Results and findings

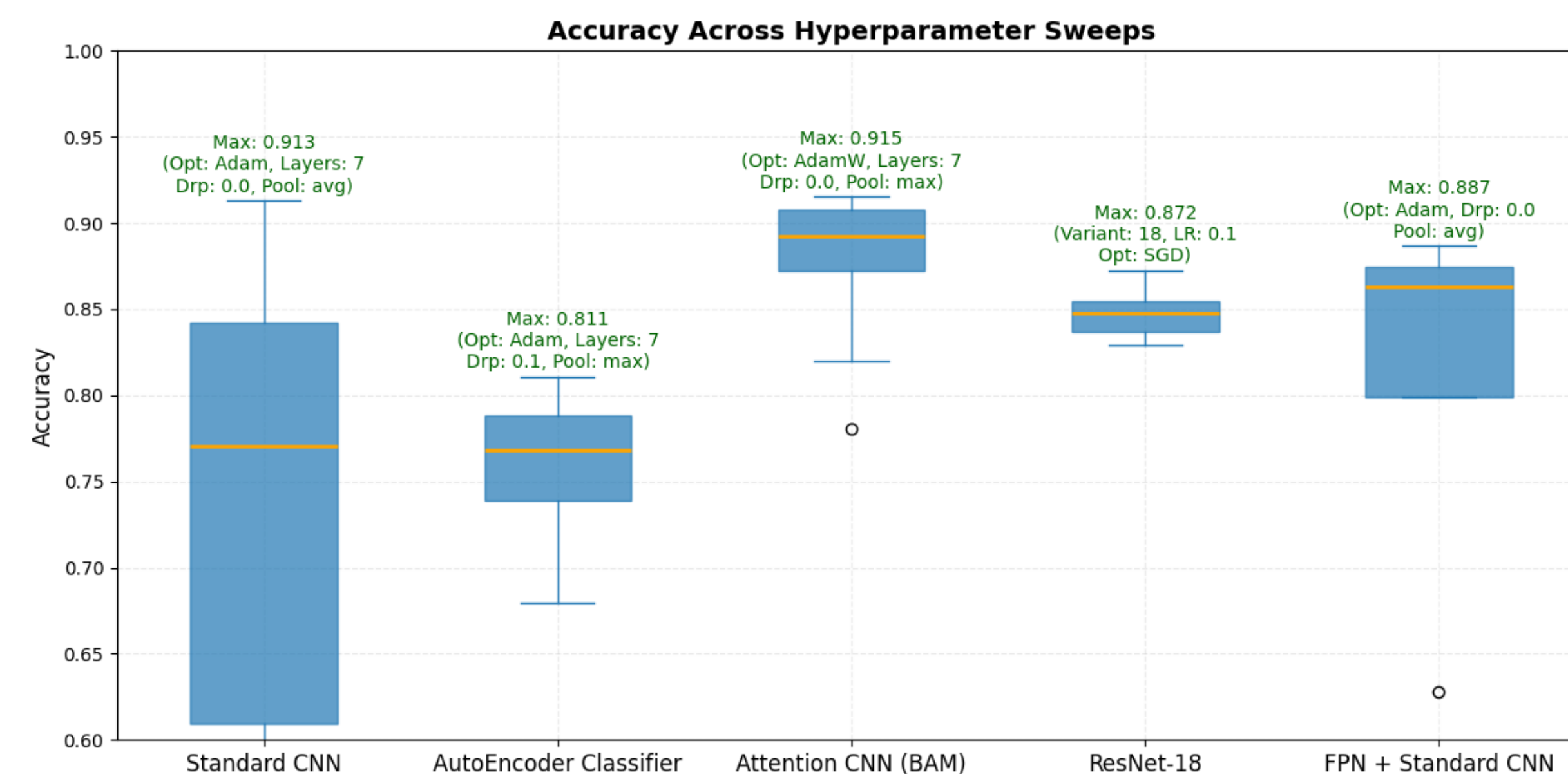


Figure 2. Classical model performance distribution

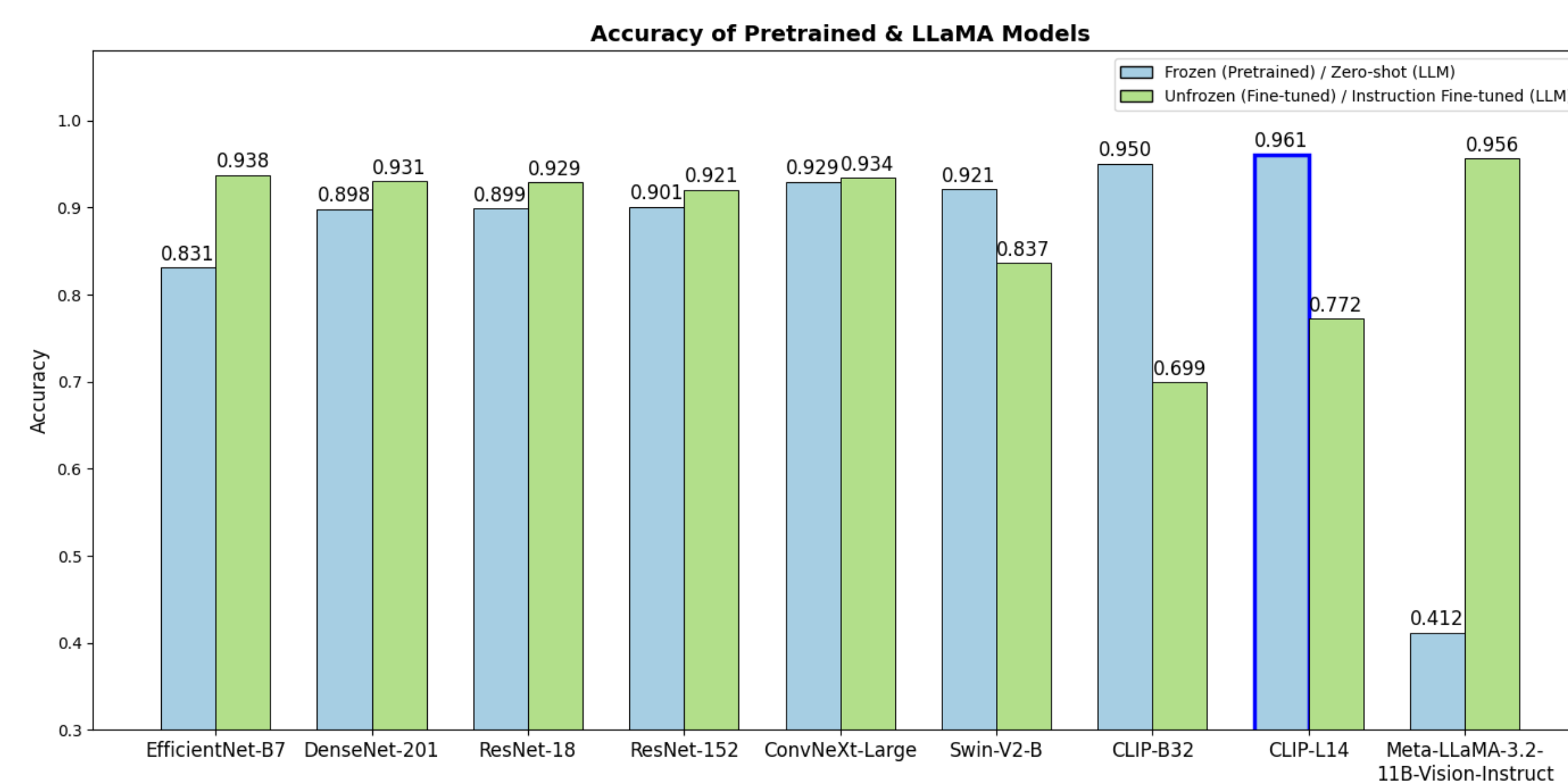
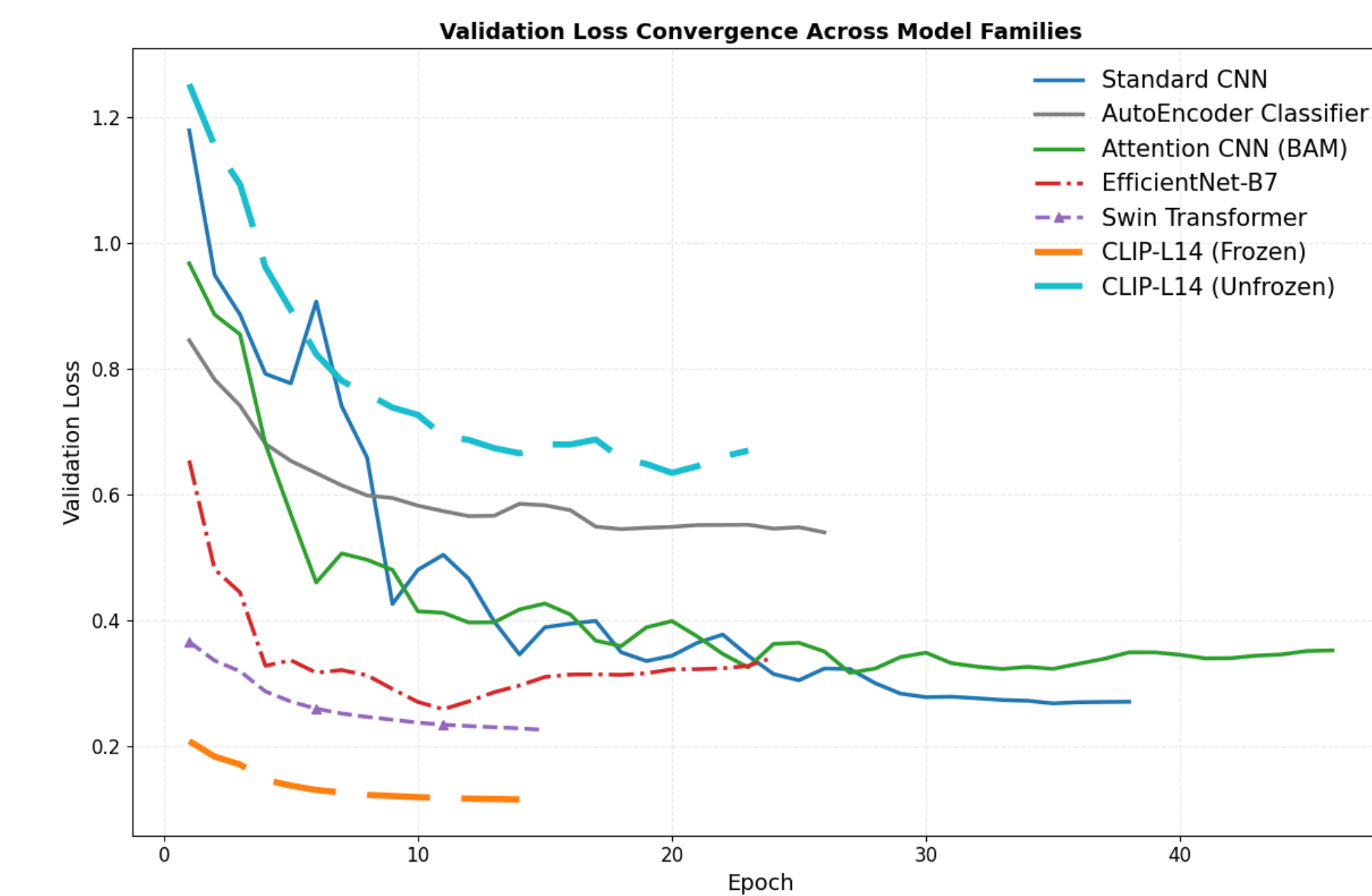


Figure 3. Frozen vs full finetuning comparison across families

Key Takeaways

- Classical CNNs show high potential but extreme variability, with performance ranging **20%-91%**, highly sensitive to architecture and hyperparameters.
- Pretrained CNNs consistently improve when unfrozen, with **EfficientNet/DenseNet/ResNet/ConvNeXt** gaining even up to **10%**.
- Pretrained Transformers (**Swin, CLIP**) degrade when unfrozen, dropping **20-25%**, showing unstable behavior.
- Finetuned **LLaMA-3.2** reaches **95.6%** on images using task-specific instruction prompts, **outperforming all unfrozen vision models**.
- Core Pattern: **CNNs benefit from adaptation, Transformers collapse when unfrozen, and LLMs gain performance** after task-specific finetuning.



- CLIP Unfrozen:** lightning-fast convergence.
- Swin:** smooth, stable descent.
- EfficientNet:** sharp early drop, then rise.
- Classical CNNs:** noisy but steady (except AutoEncoder).
- CLIP Unfrozen:** unstable, early plateau.

Figure 4. Loss curves for baseline, best classical, and pretrained models

Conclusion and future recommendations

Conclusion: Pre-trained models, especially CLIP variant, significantly outperformed all classical CNN architectures on the Intel dataset. Transfer learning proved more effective than increasing model depth, highlighting the importance of strong pretrained representations.

Future recommendations:

- Apply more advanced fine-tuning strategies (e.g., gradual unfreezing, adapters).
- Explore feature-space visualizations to better understand model behavior.
- Test lightweight ensemble methods to boost robustness.

Acknowledgments

We would like to acknowledge the following contributions and resources that made this project possible:

- Dataset:** Puneet Bansal for providing the Intel Image Classification Dataset via Kaggle.
- Course:** COMP 8740 at the University of Windsor for providing the framework and guidance for this research.